Introduction to Information Systems

PRINCIPLES OF INFORMATION SYSTEMS

Learning Objectives

- By the end of this lecture, you should be able to:
- Define what an Information System (IS) is.
- Identify the components of an IS.
- Understand how IS support business operations and decision-making.
- Recognize different types of information systems.

Data, Information, and Knowledge

- Data:
  - Raw facts
- Information:
  - Collection of facts organized in such a way that they have value beyond the facts themselves
- Process:
  - Set of logically related tasks
- Knowledge:
  - Awareness and understanding of a set of information

What is an Information System?

- **Definition:**
  An Information System (IS) is a set of interrelated components that collect, process, store, and distribute information to support decision-making and control in an organization.

- **Key Idea:**
  Information systems convert *data* into *useful information*.

Data vs Information

Term	Definition	Example
Data	Raw facts, figures	500, 2025, “Egypt”
Information	Processed data that is meaningful	“Sales in Sudan increased by 10% in 2025.”

The Process of Transforming data into Information

CHARACTERISTICS OF INFORMATION

- **If an organization's information is not accurate or complete:**
  - People can make poor decisions, costing thousands, or even millions, of dollars
- **Depending on the type of data you need:**
  - Some characteristics become more important than others
  - The Characteristics of Valuable Information

The Value of Information

- **Directly linked to how it helps decision makers achieve their organization's goals**
- **Valuable information:**
  - Can help people and their organizations perform tasks more efficiently and effectively

Components of an Information System

- **People** – users, IT staff, managers
- **Hardware** – physical devices
- **Software** – programs and applications
- **Data** – facts stored and processed
- **Procedures** – rules and guidelines
- **Networks** – communication systems

Types of Information Systems

- **Transaction Processing Systems (TPS)** – record daily operations
- **Management Information Systems (MIS)** – summarize data for managers
- **Decision Support Systems (DSS)** – assist in problem-solving
- **Executive Information Systems (EIS)** – provide top-level summaries
- **Expert Systems** – mimic human expertise

Information Systems in Business

- Support day-to-day business processes
- Enable communication and collaboration
- Provide data for analysis and decision-making
- Improve efficiency and productivity
- Create competitive advantages

Information Systems in Daily Life

- Examples:
  - Online banking
  - E-learning platforms
  - E-commerce sites
  - Social media
  - Mobile applications

The Role of IS in Organizations

- **Operational Level:** Automates routine tasks
- **Managerial Level:** Supports planning and monitoring
- **Strategic Level:** Provides insights for long-term goals

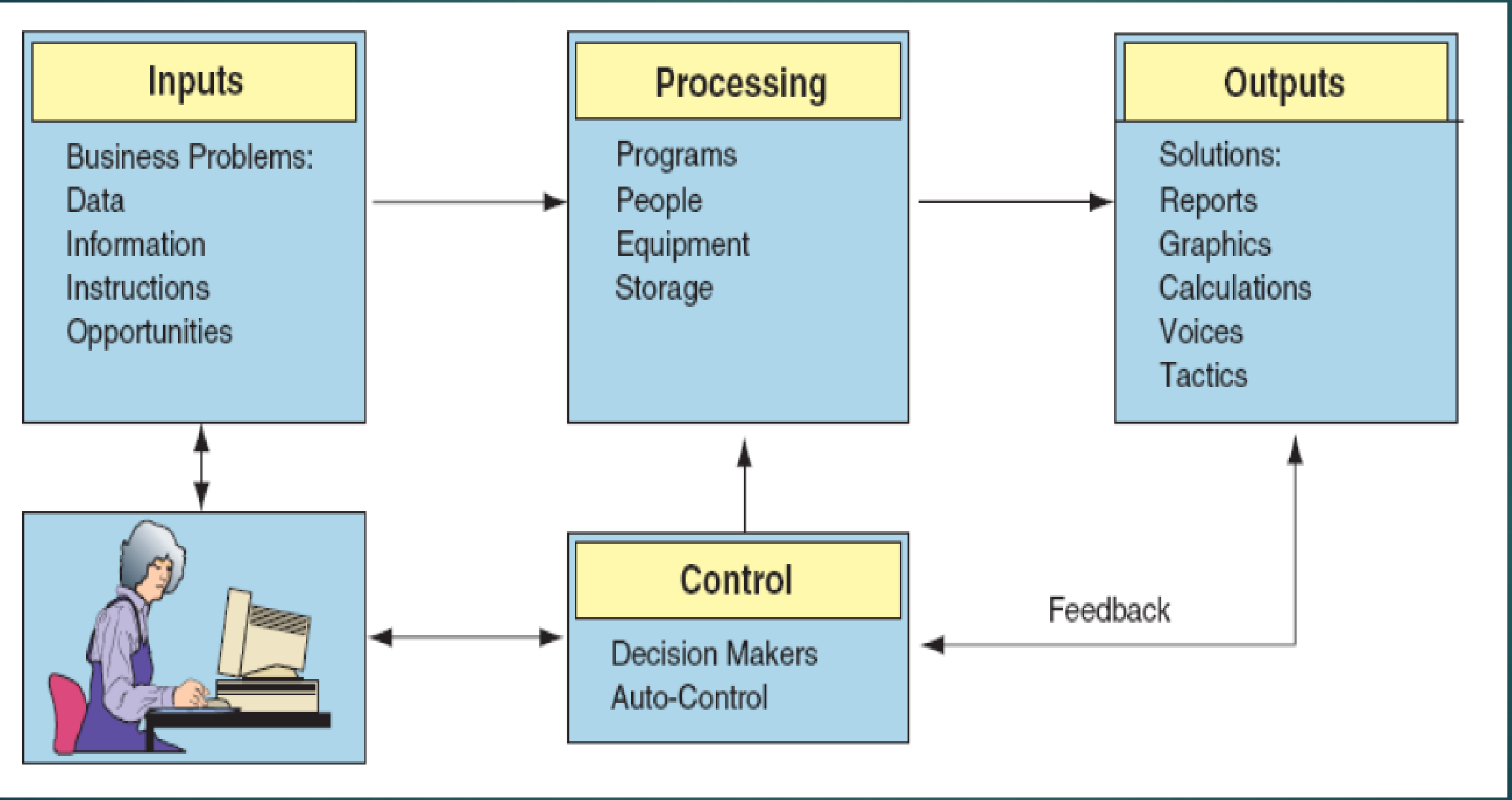
Information System vs Information Technology

Information System (IS)	Information Technology (IT)
Broader concept (includes people, data, and processes)	Focuses on hardware and software
Manages and uses information	Tools that enable IS
Example: Payroll system	Example: Computer and software used in payroll

Benefits of Using Information Systems

- Improved decision-making
- Better communication
- Increased efficiency and productivity
- Cost reduction
- Enhanced customer service

Functions of an information system

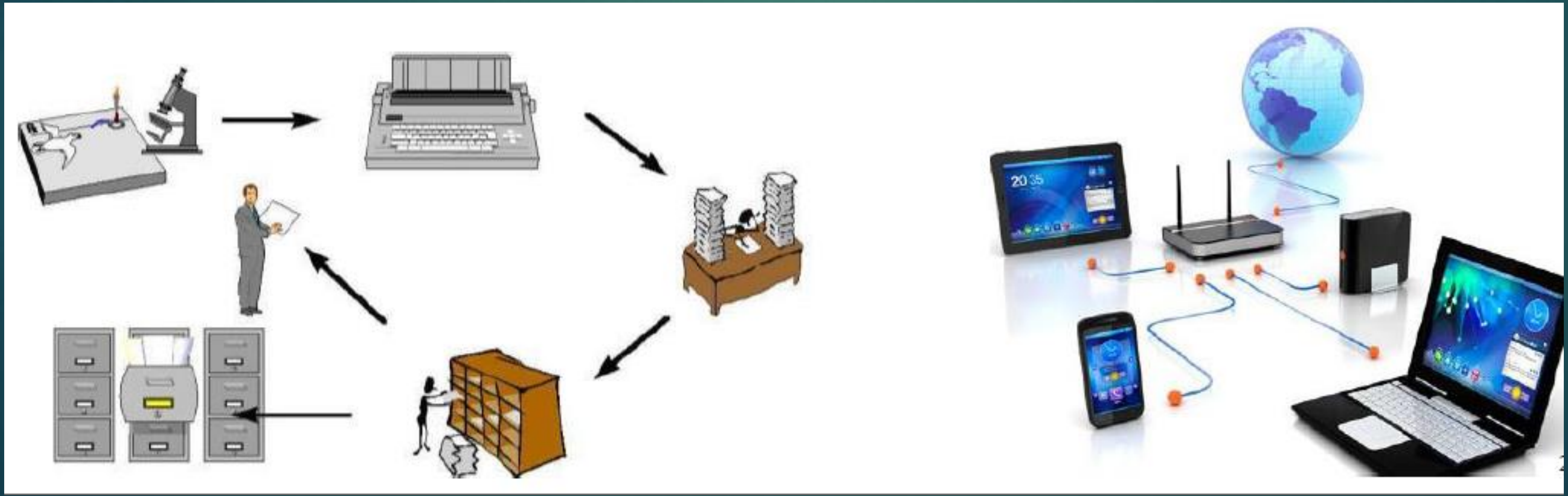


Functions of an information system

- **Input**
  - Activity of gathering and capturing raw data
- **Processing**
  - Converting data into useful outputs
- **Output**
  - Production of useful information, usually in the form of documents and reports
- **Feedback**
  - Information from the system that is used to make changes to input or processing activities

Manual and Computer-based Information Systems

- **An information system can be:**
  - Manual or computerized



A Computer-Based Information System

- A computer-based information system (CBIS) is an information system that uses computer technology to perform some or all of its intended tasks.

Challenges in Information Systems

- Security and privacy issues
- System failures
- Data quality and integrity
- Ethical concerns
- Rapid technological changes

Emerging Trends

- Artificial Intelligence (AI)
- Big Data and Analytics
- Cloud Computing
- Internet of Things (IoT)
- Blockchain Technology

Summary

- IS = People + Technology + Processes + Data
- Supports all levels of an organization
- Critical for success in the digital age

Questions

- How do information systems help organizations compete?
- What are some real-world examples of IS you use daily?
- What risks can arise from overreliance on information systems?

Information Systems in Organizations

Learning Objectives

- By the end of this lecture, students should be able to:
- Explain how information systems support organizations.
- Identify the major types of information systems used in organizations.
- Understand how IS enhance efficiency and decision-making.
- Recognize how IS contribute to organizational success and competitiveness.

What Is an Organization?

- A structured group of people working together to achieve goals.
- Organizations transform inputs (resources) into outputs (goods/services).
- Information Systems help coordinate, control, and improve this process.

Relationship Between IS and Organizations

- IS and organizations influence each other.
- IS support business processes, culture, and structure.
- Organizations determine how IS are designed and used.

Business Processes

- **Definition:** A business process is a set of related activities that produce a product or service.
- **Examples:**
  - Processing customer orders
  - Paying employees
  - Managing inventory
- Information Systems **automate**, **streamline**, and **integrate** these processes.

Functional Areas of an Organization

Department	Main Function	Example IS Used
Accounting	Record transactions	Accounting Information System
Finance	Manage budgets/investments	Financial Management System
Marketing	Promote products	Marketing Information System
HR	Manage employees	Human Resource Information System
Operations	Produce goods/services	Production Information System

Cross-Functional Information Systems

- Support multiple departments.
- Integrate data and processes across the organization.
- Examples:
  - **ERP (Enterprise Resource Planning)**
  - **CRM (Customer Relationship Management)**
  - **SCM (Supply Chain Management)**

Levels of Management

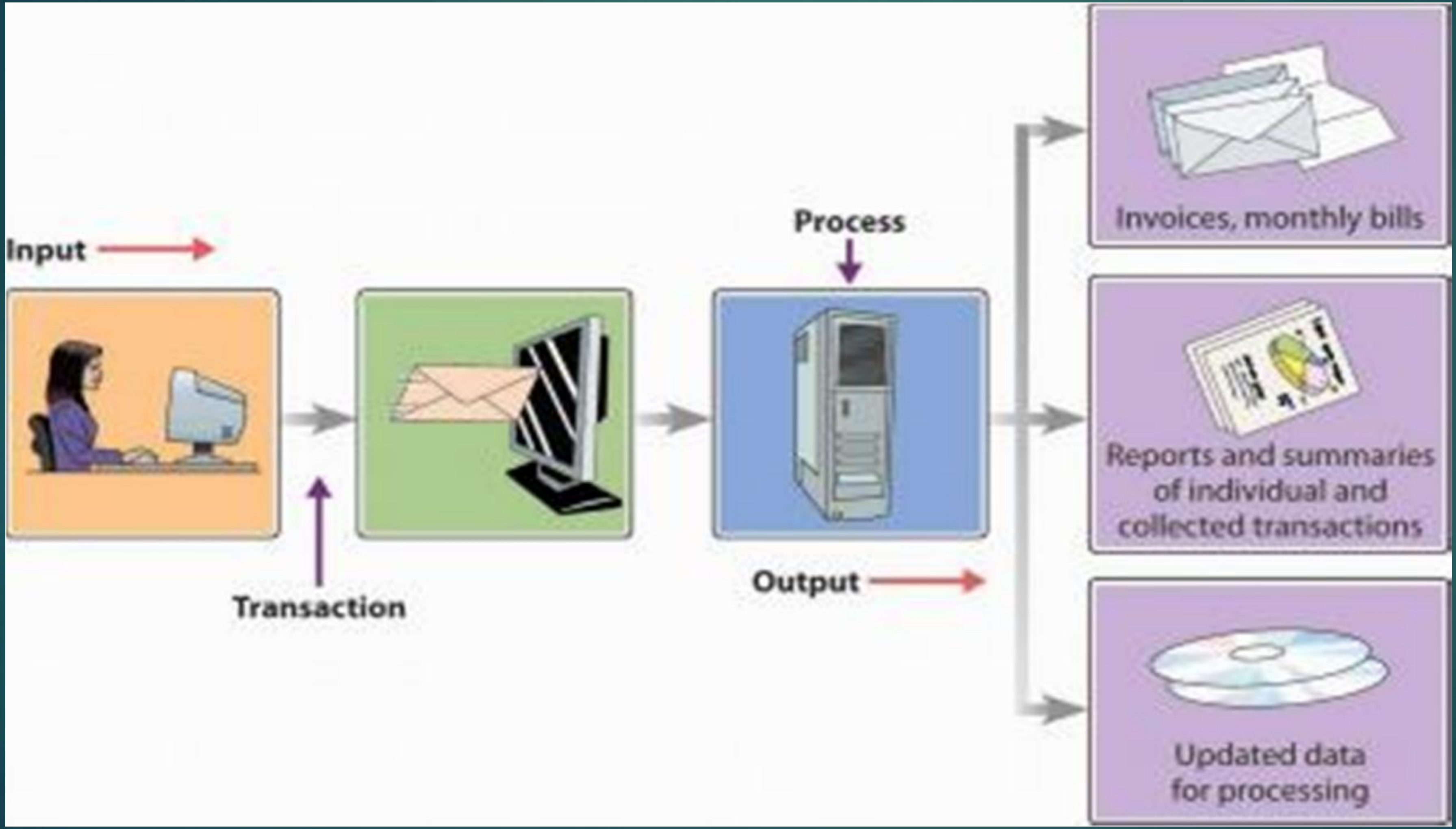
Levels of Management and IS

Level	Type of IS	Example
Operational	Transaction Processing System (TPS)	Sales order entry
Managerial	Management Information System (MIS)	Monthly sales reports
Strategic	Executive Information System (EIS)	Market trend dashboards

TRANSACTION PROCESSINGSYSTEM

- Transaction = an event that generates or modifies data
  - Used at Operational level of the organization
- Processes business events and transactions to produce reports
- Goal: to automate repetitive information processing activities
- Supports the monitoring, collection, storage, processing, and dissemination of the organization's basic business transactions
- Mainly includes accounting and financial transactions
- Mainly used for providing other information systems with data.

TPS



Input
Transaction
Process
Output
Invoices, monthly bills
Reports and summaries of individual and collected transactions
Updated data for processing

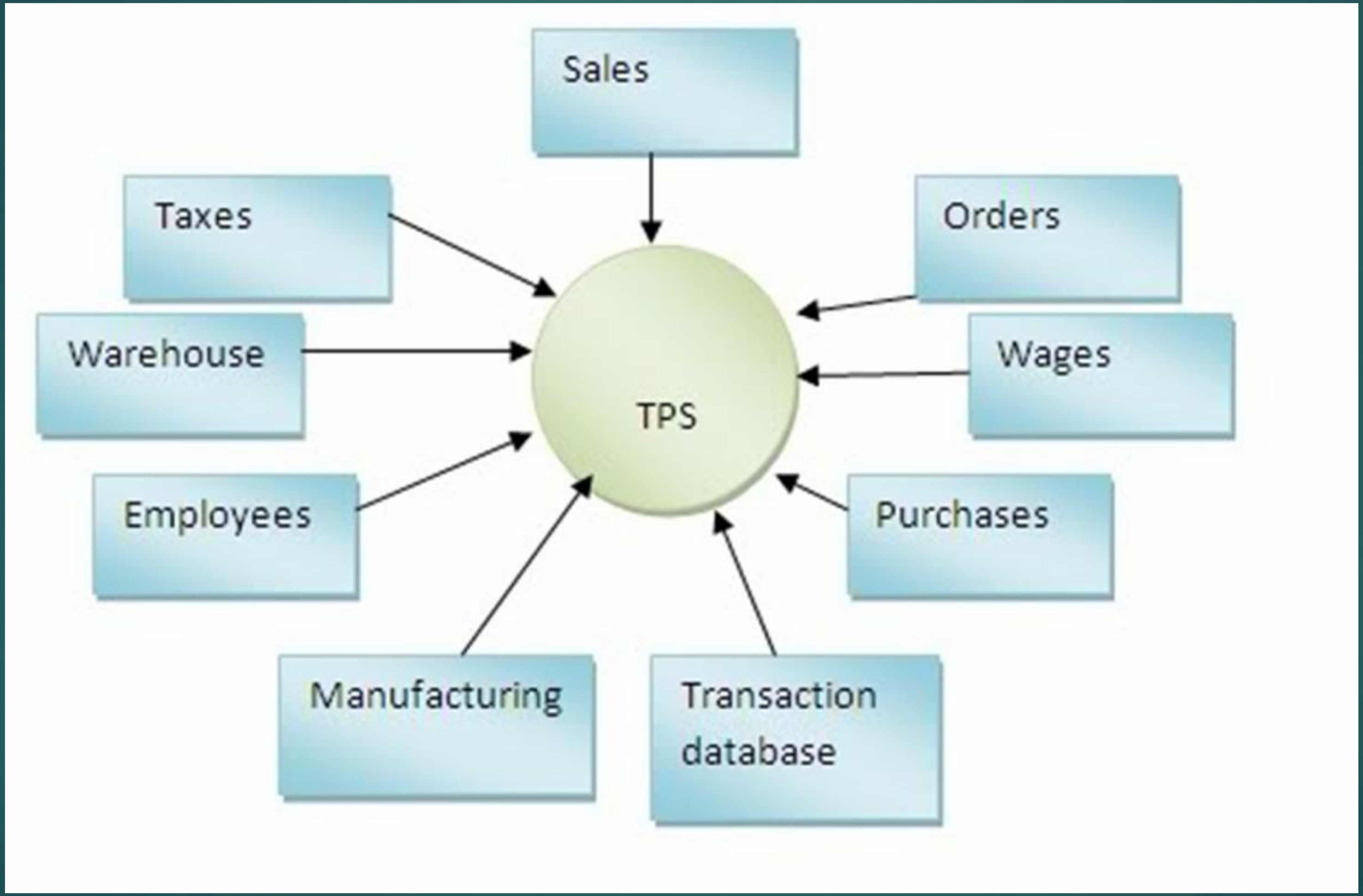
TRANSACTION PROCESSING SYSTEM

- Role of TPS
  - Produce information for other systems
  - Cross boundaries (internal and external)
  - Used by operational personnel + supervisory levels
  - Efficiency oriented
- Examples
  - Payroll processing
  - Sales and order processing
  - Inventory management
  - Accounts payable and receivable

OBJECTIVES OF TPS

- Efficient and effective operation of the organization
- Provide timely documents and reports
- Increases the competitive advantage
- Provides necessary data for tactical and strategic systems such as DSS
- Provide a framework for analyzing an organization’s activities

TRANSACTION PROCESSING SYSTEM



MANAGEMENT INFORMATION SYSTEM

- Many of the different types of Information Systems found in commercial organizations are referred to as "Management Information Systems".

- Management Information Systems are management-level systems that are used by middle managers to help ensure the smooth running of the organization in the short to medium term.

- The highly structured information provided by these systems allows managers to evaluate an organization's performance by comparing current with previous outputs.

MANAGEMENT INFORMATION SYSTEM

- Refers to the data, equipment and computer programs that are used to develop information for managerial use
- Converts raw data from transaction processing system into meaningful form
- Focus on the information requirements of low to middle level managers

MANAGEMENT INFORMATION SYSTEM

- **Role of MIS**
  - Based on internal information flows
  - Support relatively structured decisions
  - Inflexible and have little analytical capacity
  - Used by lower and middle managerial levels
  - Deals with the past and present rather than the future
  - Efficiency oriented
- **Some examples of MIS**
  - Sales management systems
  - Inventory control systems
  - Budgeting systems
  - Management Reporting Systems (MRS)
  - Personnel (HRM) systems

OBJECTIVES OF MIS

- Provide summary information of organizational activity at periodical intervals
- Operational control and efficiency
- Focus on internal information
- Useful to structured decisions

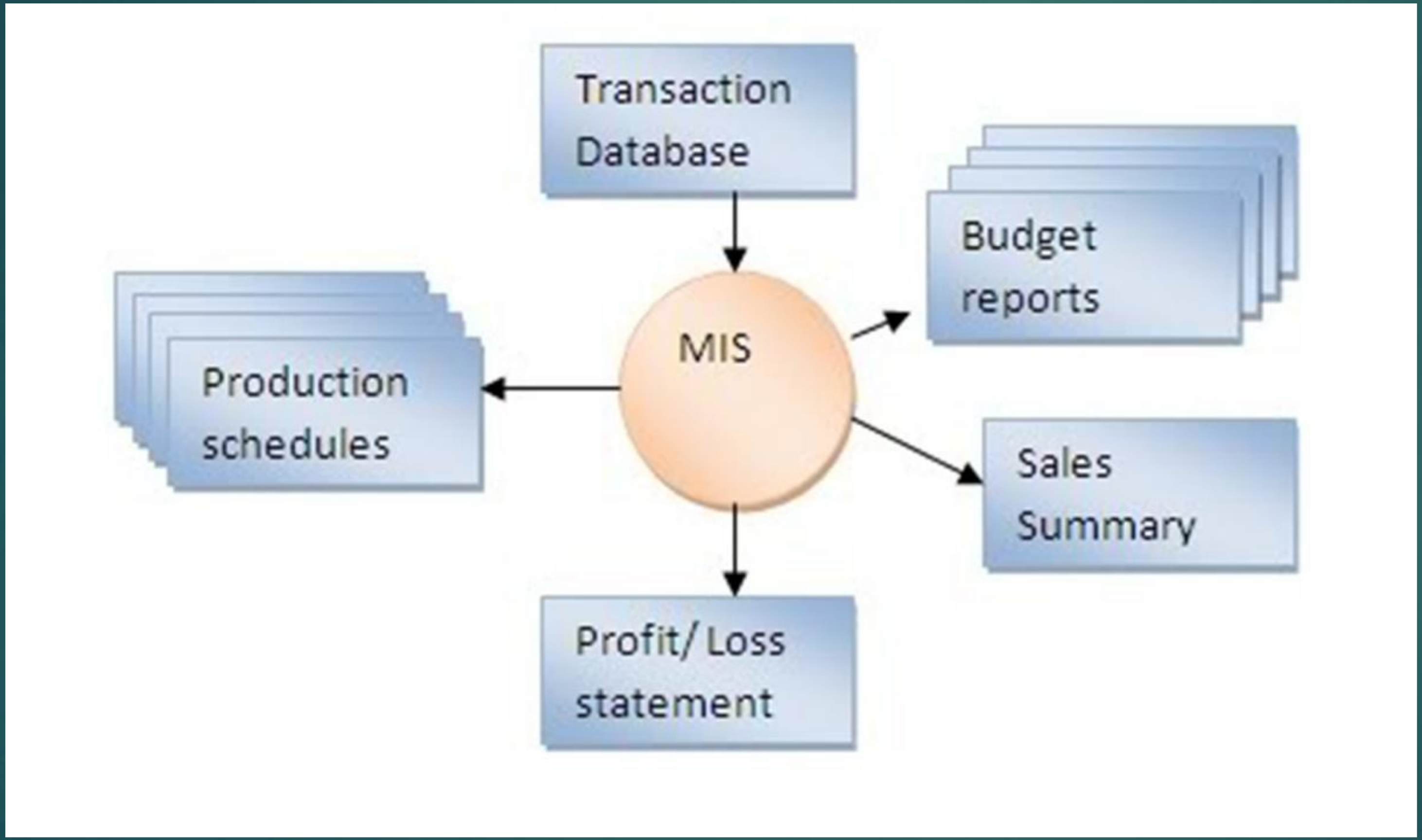
FUNCTIONS OF MIS

FUNCTIONS OF A MIS IN TERMS OF DATA PROCESSING REQUIREMENTS

INPUTS	PROCESSING	OUTPUTS
Internal transactions Internal files Structured data	Sorting Merging Summarizing	Summary reports Action reports Detailed reports

MIS are built on the data provided by the TPS

MIS


Transaction
Database
MIS
Budget
reports
Production
schedules
Sales
Summary
Profit/ Loss
statement

EXECUTIVE INFORMATION SYSTEM

- Executive Information Systems are strategic-level information systems that are found at the top of the Pyramid.
- They help executives and senior managers analyze the environment in which the organization operates, to identify long-term trends, and to plan appropriate courses of action.
- The information in such systems is often weakly structured and comes from both internal and external sources.
- Executive Information System are designed to be operated directly by executives without the need for intermediaries and easily tailored to the preferences of the individual using them.

EXECUTIVE INFORMATION SYSTEM

- Specialized form of DSS
- Used by top-level managers
- Reduce the information overload on executives
- Makes use of internal and external information
- Provides managers and executives flexible access to information for monitoring operational results and general business conditions
- Provides a comprehensive picture of business performance by analysing key performance indicators for growth
- Meets strategic information needs of the top management
- Also known as Executive Support System

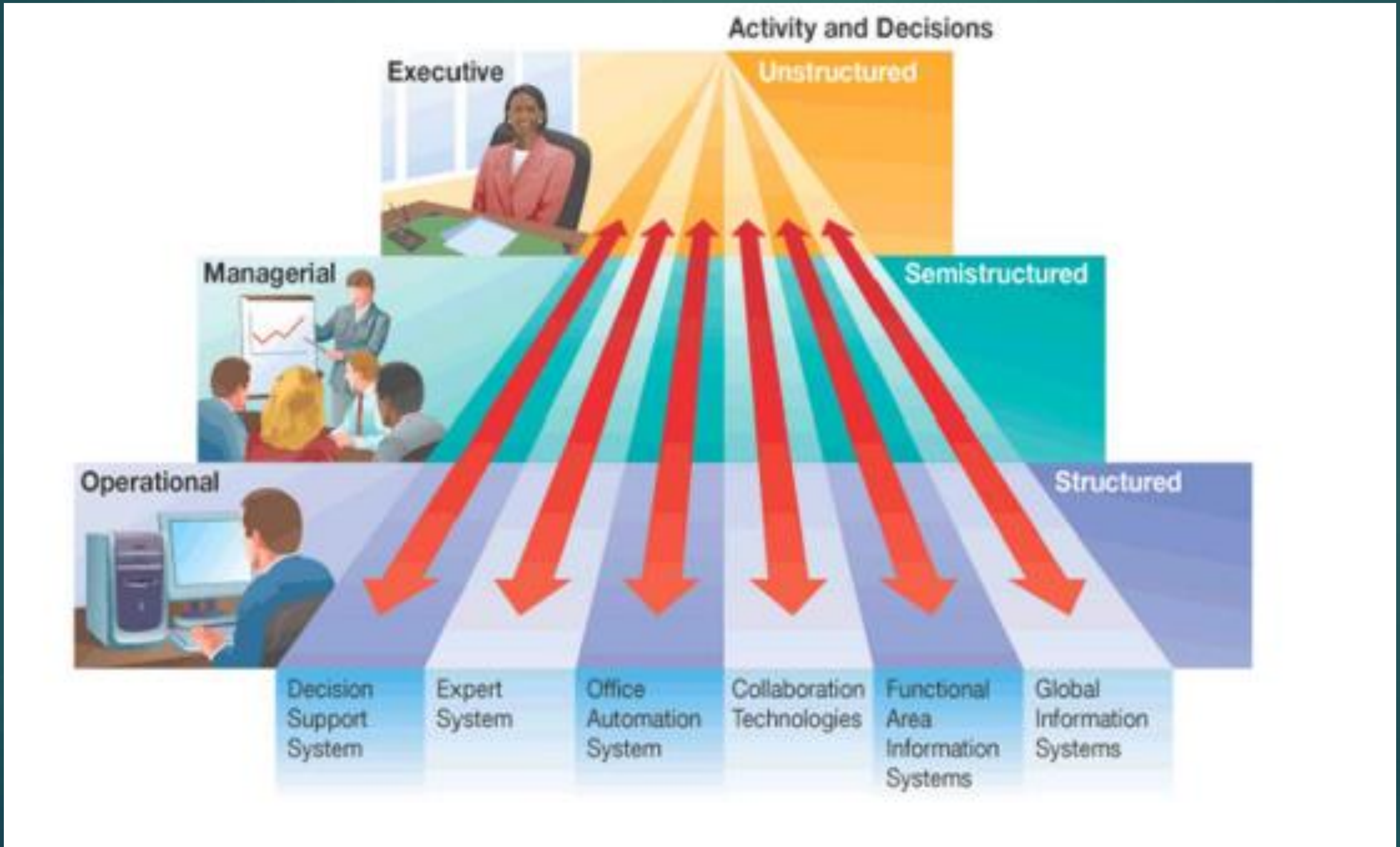
EXECUTIVE INFORMATION SYSTEM

- Role of EIS
  - Are concerned with ease of use
  - Are concerned with predicting the future
  - Are effectiveness oriented
  - Are highly flexible
  - Support unstructured decisions
  - Use internal and external data sources
  - Used only at the most senior management levels
- Some examples:
  - Executive Information Systems tend to be highly individualized and are often custom made for a particular client group; however, a number of off-the-shelf EIS packages do exist and many enterprise level systems offer a customizable EIS module

FUNCTIONS OF AN EIS

INPUTS	PROCESSING	OUTPUTS
External Data Internal Files Pre-defined models	Summarizing Simulation "Drilling Down"	Summary reports Forecasts Graphs / Plots

Information Systems that Span Organizational Boundaries



Decision-Making Levels of an Organization

- **Executive level (top)**
  - Long-term decisions
  - Unstructured decisions
- **Managerial level (middle)**
  - Decisions covering weeks and months
  - Semi-structured decisions
- **Operational level (bottom)**
  - Day-to-day decisions
  - Structured decisions

Information Flow in Organizations

- **Vertical Flow:** Between top, middle, and lower management.
- **Horizontal Flow:** Between departments or teams.
- **External Flow:** Between organization and environment (customers, suppliers, etc.).

Examples of Information Systems by Function

- **HR System:** Tracks employee records and performance.
- **Inventory System:** Monitors stock levels and orders.
- **Accounting System:** Manages financial data and reports.
- **Customer Relationship System:** Tracks customer interactions.

How IS Add Value

- Improves efficiency and accuracy
- Enhances communication and collaboration
- Supports better decision-making
- Reduces costs
- Improves customer satisfaction
- Creates competitive advantages

Challenges of Implementing IS

- Resistance to change
- High implementation costs
- Data security and privacy issues
- Need for continuous updates and training

Summary

- IS are essential tools that support all organizational levels.
- They integrate people, processes, and technology.
- Effective IS use leads to improved performance and competitiveness.

Questions

- Why is it important for different departments to share information?
- How can IS improving communication within an organization?
- What problems might occur if an IS fails?